

# Abhinav Karnatak

Full-Stack Developer | Applied AI & Agentic Engineering

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**Full-stack developer** focused on building end-to-end applications with **React/Next.js, Node.js, and microservices architecture**, with a growing focus on **AI and agentic engineering**. Building hands-on projects with **LangChain, LangGraph, RAG, and LLM integrations**, combining AI capabilities with practical full-stack systems.

## SKILLS

**AI/LLM Engineering:** LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Agentic Workflows, LLM Evaluation, Guardrails, Prompt Engineering, LLM API Integration

**Programming Languages:** JavaScript, TypeScript, C++, Python

**Frontend Technologies:** React.js, Next.js, React Native, Tailwind CSS, Framer Motion, Progressive Web Apps (PWA), Figma

**Backend & Databases:** Node.js, Express.js, MongoDB, PostgreSQL, Prisma, Redis, RabbitMQ

**DevOps and Cloud:** Docker, AWS (EC2, S3), Nginx

**Libraries & Integrations:** Socket.io, Gemini API, Razorpay, Scrapy, Recharts

## PROJECTS

### Nimbus: Autonomous Cloud Coding Agent

- Built an **autonomous cloud coding agent** using **TypeScript, Node.js, React, MongoDB, Redis, and LangGraph** that connects to **GitHub repositories**, understands and navigates codebases, answers repository-level queries, and autonomously works on coding tasks.
- Engineered an **agentic workflow with LangGraph** for iterative **LLM reasoning, codebase retrieval, and tool execution**, enabling the agent to inspect files, modify code, execute commands, and validate changes through **build, lint, and test checks**.
- Designed a secure **E2B sandboxed execution architecture** where model-proposed commands and code changes execute in an **isolated environment** without sensitive credentials, while the trusted backend validates resulting patches and handles authenticated **GitHub operations and pull requests**.

### YapIt: Real-Time Microservices Chat Application

- Built using **Next.js, Node.js, Socket.io, and Redis**, YapIt is a highly scalable full-stack web app where users can engage in seamless real-time conversations.
- Engineered a decoupled **Microservices Architecture** (User, Chat, and Mail services) utilizing **RabbitMQ** for reliable inter-service messaging and asynchronous task handling like global account data cleanup.
- Containerized the entire ecosystem with **Docker Compose** and deployed it to an **AWS EC2** instance, utilizing **Nginx** reverse proxy to securely manage API routing, continuous WebSocket traffic, and AWS S3 file-sharing.

### Revuea: Empowering Honest Feedback with Anonymity

- Developed a full-stack web application using **React.js, Node.js, and PostgreSQL** (via **Prisma ORM**) to securely collect, store, and manage anonymous user feedback.
- Implemented a secure authentication flow using **OTP-based signups** and built a dashboard utilizing **Recharts** to visualize feedback data efficiently.
- Integrated the **Gemini AI API** to instantly process large volumes of user feedback into readable text summaries, while also adding **CSV export** for external data analysis.

### PixScribe: SaaS Text-to-Image Generator Web App

- Built using **React.js, Node.js, Express, and MongoDB**, PixScribe is a full-stack web app that lets users generate AI-based images from text prompts.
- Engineered a credit system linked to a **Razorpay** integration, handling secure payment processing and automatically updating user account balances upon successful transactions.
- Designed a fast, highly responsive user interface with **Tailwind CSS** and integrated **Framer Motion** to deliver smooth layout animations and state transitions.

## WORK EXPERIENCE

### Analyst (Software Engineer)

Capgemini

Sep 2025 - May 2026

Mumbai, Maharashtra

- Assisted in troubleshooting network and connectivity issues, gaining hands-on exposure to enterprise server environments (Linux/Windows) while ensuring timely resolution of assigned tickets.
- Collaborated with the team to monitor system health and ensure smooth daily operations.

### Data Engineer Intern

HERE Technologies

Jun 2024 - Jul 2024

Remote

- Built custom Python web crawlers using **Scrapy** to scrape store locations from retail websites with varying HTML layouts, scheduling automated runs that cut manual effort by 30%.
- Documented the workflow and codebase to make it easier to maintain and update in the future.

## EDUCATION

### B.Tech Computer Science and Engineering

University Of Petroleum and Energy Studies

Aug 2021 - May 2025

- Served as the **Student Placement Representative (SPR)**, coordinating communication between the placement cell and students during campus recruitment activities.